Latest feature-engineering layout used by PLC models

Protein complete V4

62D physchem:

AA/v2 descriptors, sequence, hbond, residue ID, SS3, class

89D spatial scalars:

SASA/surface/geometry, pocketness, torsion, packing, charge/hydrophobicity, metal

8x3 vectors:

sidechain, aromatic, H-bond, cavity directions

Ligand V5

109D per atom:

local54 + global45 + shape10

local54 chemistry:

atom identity/topology, charge, LogP/TPSA, RMSF, SASA, HBD/HBA, rotatable, metal, stereo

graph/geometry:

6D bond edge_attr + coords

Current integration

Protein trunk:

V4 features + ESM2, optional weak ESM3 prior

Ligand path:

late local_delta K/V from local54 and ligand graph

V11m13 options:

local54 group adapter, residue-chem compat, edge-conditioned K/V